

Interactive Visual Storytelling for Social Impact

You will work in groups (the same group as Assignment 1) to design and implement an **interactive visual storytelling** on a topic with the potential **social impact** (for the good of society).

You are free to choose any topic that you are excited about, including (but not limited to): *healthcare, environment, mobility, inequality, education, public spending, energy, and housing*. The goal is not only to visualize data, but also to **communicate insights, support exploration, and tell a meaningful story** that informs or empowers the public.

Assignment 2: Project Proposal

Due: Monday, 11 May 2026, 23:59 CET

Before starting the final project, each team will submit a project proposal report describing the socially relevant topic you select, the dataset you plan to use, and the target audience you aim to communicate with. Based on these elements, you will outline the story or narrative you want to explore and develop a design sketch for your interactive visual storytelling project.

Part 1: Project Description

In the first part, prepare a short report (around 1-2 pages) describing your project.

1. Selected Topic and Motivation

Select a topic with potential social impact, such as healthcare, environment, mobility, inequality, education, public spending, energy, or housing.

Clearly state the question you want to explore in your final project. Explain why this question is important and what societal issue it addresses. Your motivation may be supported by references to news articles, reports, or other background sources, or by real-world examples that illustrate why the topic is relevant.

2. Dataset

Identify one or more datasets related to your topic. You may obtain data from open data platforms (e.g., [Offene Daten Österreich](#)) or other comparable public datasets. The dataset should be sufficiently rich and complex to support multiple visualizations and perspectives in your final project.

In the report, provide a description of the dataset(s), including: 1) what the data contains, 2) where the data comes from, and 3) a URL link to access the dataset.

3. Target Audience

Define the target audience for your visualization. Examples may include citizens, policymakers, journalists, students, researchers, or NGOs.

What is the purpose or goal of your narrative visualization? For example, are you trying to help policymakers identify issues that need attention? Are you aiming to raise awareness among the general public? Or are you making a case to a specific group of stakeholders (e.g., suburban homeowners)?

Clearly defining your audience will help determine how the data should be framed and communicated. For instance, a specialized audience may allow for more detailed analysis, while a broader audience may require more contextual explanation.

4. Main Question, Storytelling, or Narrative

Describe the insight, narrative, or message you want to communicate through your visualization. Explain how the dataset may help answer your question and what kind of story or exploration the visualization might enable.

Part 2: Design Sketches

In the second part, prepare a **visualization sketch, wireframe, or storyboard** that shows how your interactive visual storytelling project might look.

Think of your sketch as a visual outline of your project. It should communicate:

- The overall narrative structure
- The visual layout of the interface
- The interactive flow of your visual storytelling

In your sketch, you should indicate how many pages or views your project will include, what types of visualizations appear on each view, how they are arranged, and how users navigate between them. Also consider which visualizations are interactive and which are static, and whether you plan to use interactions or animations to support engagement and understanding.

When creating your wireframes, design from your target audience's perspective. Think about the key messages or insights you want them to take away, and how each step of the story guides their attention and exploration. Use narrative elements such as highlights, annotations, or callouts to emphasize important points and provide context throughout the experience.

You are free to use **any tool or template** to create your design sketch. There are no restrictions on format or style. The sketch does not need to be polished; clarity of your ideas and storytelling is what matters most. Simple hand-drawn sketches are preferred, but digital sketches or rough mockups are fine as well.

Finally, along with your sketch, **write a 1-2 paragraph explaining the design rationale** on your final sketch, including:

- Why did you choose these particular visualizations and layouts?
- How do they support your story and the users' tasks?
- How will users explore and make sense of the data through your design?

Submission

Upload your proposal as a PDF file to TUWEL, including your project proposal and design sketches.

The report should not exceed 5 pages and must be **written in English**.

Use of Generative AI

The use of generative AI tools is not permitted for this assignment. The goal of this task is to help you practice and develop the fundamental skills yourself. If generative AI is used, the assignment will be **treated as plagiarism** and handled in accordance with the university's academic integrity guidelines.

Assignment 3: Project Implementation

Due: Monday, 15 June 2026, 23:59 CET

At this final stage, you will implement your project based on your proposal and design sketches. Your deliverable is an interactive visualization deployed on the website. The final implementation does not need to match your sketches exactly, but it should build on the ideas and structure you proposed. Be sure your project includes a clear storytelling component that combines narrative elements with interactive visualizations to guide users through the data and highlight key insights.

Formal Requirements

- **At least n different visualization components**, where n equals the number of team members. Multiple charts of the same type count as a single visualization; e.g., three bar charts count as a single visualization component.
- **At least three different interaction techniques**, e.g., brushing and linking, dynamic querying, input widgets, filtering, zooming, and multiple-coordinated views.
- **At least $2n$ narrative views or story components**. These may include different visualizations, transitions, or narrative steps that guide the audience through the story. Transitions, highlights, or interactive steps that reveal different insights within the same visualization can also count toward these story components.

There are no restrictions on the programming tools or libraries you may use to implement your project. You are free to choose any web-based visualization framework or technology, but are encouraged to use the libraries demonstrated in the class (i.e., D3.js). **However, the use of dashboarding or non-programming software (such as Tableau, Power BI, or similar tools) is not allowed.**

Implementation

To develop your project, **create a GitHub code repository and deploy the final visualization on the web.**

- Your web application should be stand-alone and include an `index.html` file as the entry point.

- You may either store static datasets (optionally preprocessed) in your Git repository or access live data through an API.
- Your code must be readable, reasonably structured, and consistent in style.
- Git commits should be reasonably sized and include informative commit messages.
- In the **README** file, provide readable instructions that allow others to clone the project and run the code locally, including any required dependencies, setup steps, and commands.

Presentation

Each group will register for a presentation time slot (submission talk) on TUWEL. During the presentation session, each group will give a 10-minute presentation (including Q&A) consisting of the following parts:

- **2-minute elevator pitch** briefly introducing your topic, motivation, data source, and target audience
- **5-minute project demonstration**, walking through the interactive visualization with a storytelling narrative. This part should include a live demo of possible interactions and the most interesting insights or findings from your visualization.
- **3-minute Q&A session**

All group members are expected to present and help answer the questions.

Attendance for the entire presentation session is mandatory. During the session, we may randomly select teams to ask questions about other projects. Therefore, everyone is expected to listen and actively participate.

Submissions

You will submit **your presentation slides in PDF format** on TUWEL. On the last slides, include the following information:

1. **Include the contribution statement**, describing how each team member contributed to the project (i.e., who worked on which parts).
2. **Provide the URL links** to 1) the deployed visualization website and 2) the source code repository on GitHub. Please also include these links in the comment section of your team submission on TUWEL.
3. **Disclosure statement** describing the use of generative AI tools (if applicable) in the final project implementation.

Use of Generative AI

For Assignment 3, the use of generative AI tools is allowed to **help you learn and implement technical aspects of your project**.

However, the following conditions apply:

- AI usage must be clearly documented in accordance with [TU Wien guidelines](#).
- You must explicitly state if and how AI tools were used, and indicate which parts were generated, revised, or assisted by AI.
 - Provide this disclosure statement in your presentation slides upon submitting to TUWEL.
- You must fully understand the code and be able to explain and modify it yourself.

- Overall effort will be graded. Using AI support means we expect you to invest more effort and creativity in developing an outstanding idea, thoughtful design, and clear rationale.

Resources

You may find the following websites useful as **inspiration for your project**.

- Information is Beautiful <https://informationisbeautiful.net/>
- The Pudding <https://pudding.cool/>
- Visualizations from Complexity Science Hub Vienna <https://vis.csh.ac.at/>
- New York Times <https://www.nytimes.com/spotlight/graphics> and <https://www.nytimes.com/interactive/2025/12/22/us/2025-year-in-graphics.html>

Visual Storytelling

- Storytelling and Data Visualization <https://data.europa.eu/apps/data-visualisation-guide/storytelling-and-data-visualisation>
- Communicating with Interactive Articles <https://distill.pub/2020/communicating-with-interactive-articles/>
- From Static to Stories: The Evolving Landscape of Data Visualization – Duncan Clark <https://www.youtube.com/watch?v=woODmojQdBw>
- What it takes to create visual stories at the New York Times – Marco Hernandez <https://www.youtube.com/watch?v=mpsTBRWjI6w>

Scrollytelling

- Scrollytelling: Introduction <https://data.europa.eu/apps/data-visualisation-guide/scrollytelling-introduction>
- So You Want to Build A Scroller – Jim Vallandingham <https://vallandingham.me/scroller.html>
 - Also see [his talk](#) at the OpenVis conference 2015 and [examples](#)
- How to implement scrollytelling with six different libraries – Russell Samora
- Scrollytelling libraries
 - Scrollama <https://github.com/russellsamora/scrollama>
 - Waypoints <https://github.com/imakewebthings/waypoints>